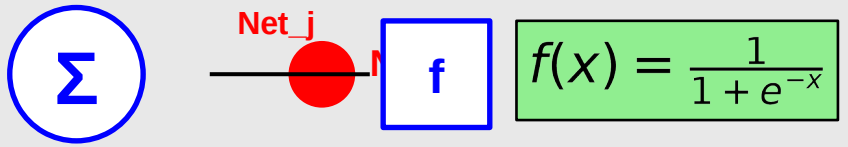
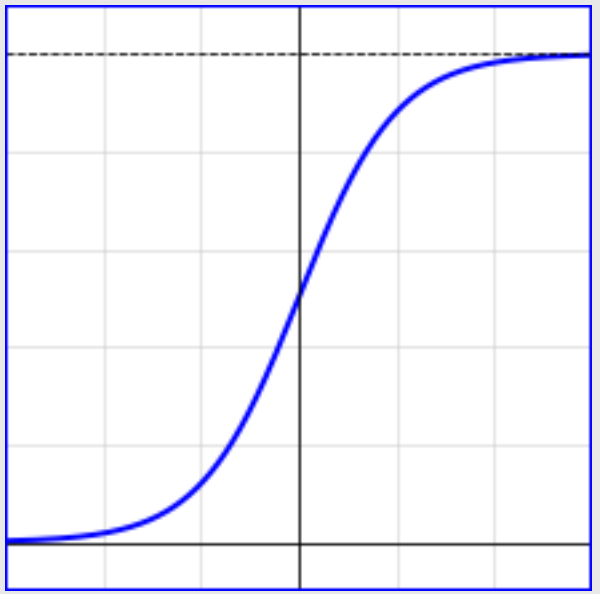
Remember: Derivative of Sigmoid Function

$$O_j = f(Net_j) = \frac{1}{1+e^{-Net_j}}$$



$$f(x) = \frac{1}{1+e^{-x}}$$



$$f'(Net_j) = \frac{e^{-Net_j}}{(1+e^{-Net_j})^2}$$

$$f'(Net_j) = \frac{1}{1+e^{-Net_j}} \cdot \frac{(1+e^{-Net_j})-1}{1+e^{-Net_j}}$$

$$f'(Net_j) = \frac{1}{1+e^{-Net_j}}\left(1 - \frac{1}{1+e^{-Net_j}}\right)$$

$$f'(Net_j) = O_j(1-O_j)$$